APPLICATIONS OF COMPUTATIONAL FLUID DYNAMICS

UNIT – II

Hyperbolic Equations

Syllabus (as given)

- Hyperbolic equations: explicit schemes and Von Neumann stability analysis, implicit schemes, multi-step methods, nonlinear problems, second-order one-dimensional wave equations.
- Burgers equations: explicit and implicit schemes, Runge–Kutta method.

1. Hyperbolic Equations — Introduction

Hyperbolic PDEs describe wave propagation and convection (transport without diffusion): sound waves, shock waves, supersonic flow, flood waves. Two basic model equations:

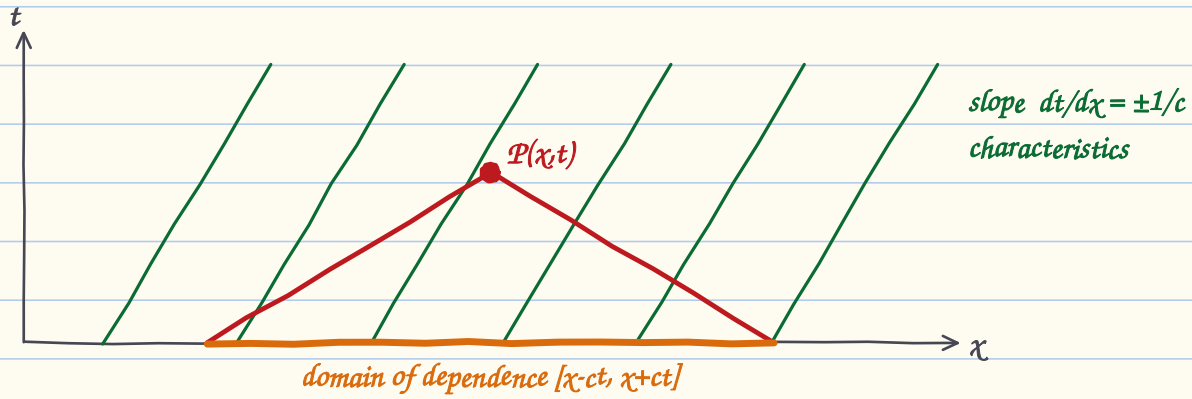
$$\text{First-order wave (linear convection):}\quad \partial u/\partial t + c\,\partial u/\partial x = 0\,,\quad c > 0$$

$$\text{Second-order wave equation:}\quad \partial^2 u/\partial t^2 = c^2\,\partial^2 u/\partial x^2$$

Exact solution of the first-order equation: $u(x,t) = f(x - ct)$ — the initial profile $f(x)$ simply translates to the right with speed c without changing shape.

Characteristics and domain of dependence

Along the lines $dx/dt = c$ (characteristics) the solution u is constant. Information travels at finite speed c along these lines.



- Value at $P(x,t)$ depends only on the initial data in the domain of dependence $[x - ct, x + ct]$ (for the 2nd-order equation); the region influenced by a point is its range of influence.
- A numerical scheme must respect this: its numerical domain of dependence must contain the

physical one -> CFL condition.

Courant–Friedrichs–Lewy (CFL) condition

$$\nu = c\,\Delta t / \Delta x \le 1$$

- ν is the Courant number (CFL number). It means that the wave must not travel more than one grid cell in one time step in an explicit scheme.
- CFL is necessary for stability of explicit schemes, but not always sufficient.

2. Explicit Schemes for $u_t + c\,u_x = 0$

Notation: $u_i^n = u(i\Delta x, n\Delta t)$, $\nu = c\Delta t/\Delta x$, Von Neumann: $u_i^n = g^n e^{I\theta}$, $G = g^{n+1}/g^n$.

(a) FTCS — forward time, central space

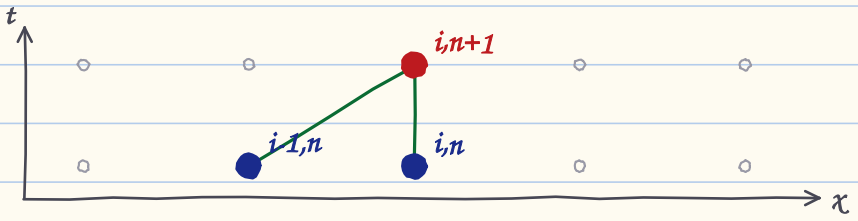
$$u_i^{n+1} = u_i^n - (\nu/2)(u_{i+1}^n - u_{i-1}^n)$$

$$G = 1 - (\nu/2)(e^{I\theta} - e^{-I\theta}) = 1 - I\nu \sin\theta \quad \rightarrow \quad |G|^2 = 1 + \nu^2 \sin^2\theta > 1$$

Unconditionally unstable for every ν — never use FTCS for pure convection.

(b) First-order upwind (FTBS for c > 0)

Upwind stencil (c > 0)



$$u_i^{n+1} = u_i^n - \nu\,(u_i^n - u_{i-1}^n)$$

$$G = 1 - \nu(1 - e^{-I\theta}) \quad \rightarrow \quad |G|^2 = 1 - 2\nu(1-\nu)(1 - \cos\theta)$$

- Stable when $0 \le \nu \le 1$. Accuracy $O(\Delta t, \Delta x)$. Strongly diffusive: modified equation $u_t + c\,u_x = (c\Delta x/2)(1-\nu)\,u_{xx}$ (numerical / artificial viscosity smears sharp fronts).
- Physical reason: information comes from upstream, so use upstream points. If $c < 0$ use $u_{i+1} - u_i$.

(c) Lax (Lax–Friedrichs) scheme

$$u_i^{n+1} = \tfrac{1}{2}(u_{i+1}^n + u_{i-1}^n) - (\nu/2)(u_{i+1}^n - u_{i-1}^n)$$

$$G = \cos\theta - I\nu \sin\theta \quad \rightarrow \quad |G|^2 = \cos^2\theta + \nu^2 \sin^2\theta \le 1 \text{ for } \nu \le 1$$

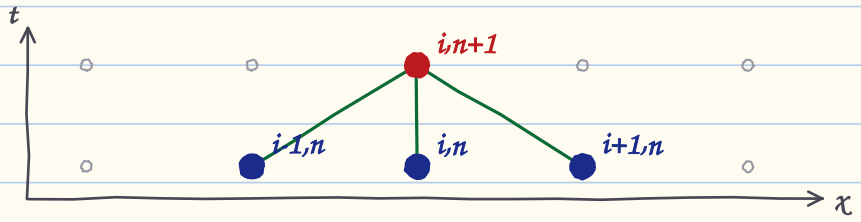
- Replaces u_i^n in FTCS by its spatial average -> stable ($\nu \le 1$), first-order, heavily dissipative.

(d) Lax–Wendroff scheme (second order)

Taylor series in time: $u^{n+1} = u^n + \Delta t\, u_t + (\Delta t^2/2)\, u_{tt}$, with $u_t = -c\, u_x$ and $u_{tt} = c^2 u_{xx}$. Central differences give:

$$u_i^{n+1} = u_i^n - (\nu/2)(u_{i+1}^n - u_{i-1}^n) + (\nu^2/2)(u_{i+1}^n - 2u_i^n + u_{i-1}^n)$$

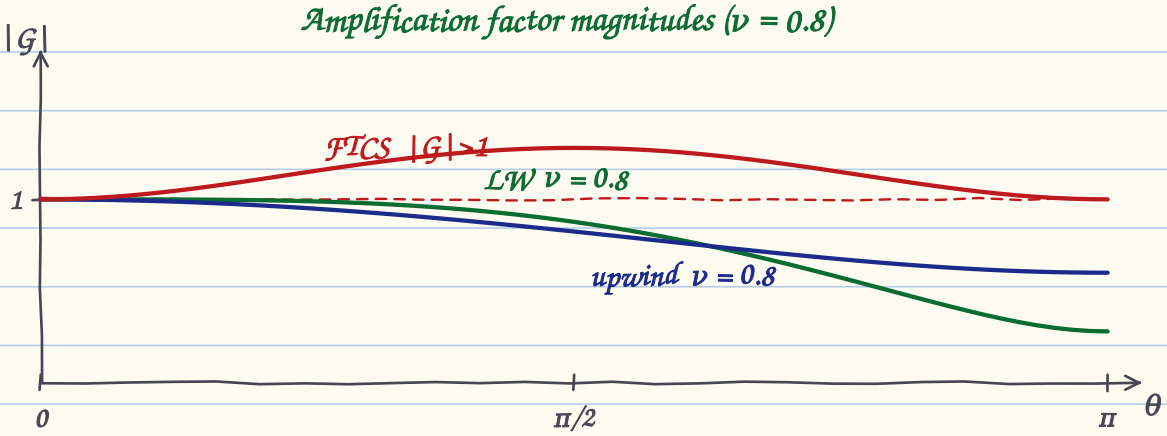
Lax–Wendroff stencil (3 known, 1 unknown)



$$G = 1 - \nu^2(1 - \cos\theta) - I\nu \sin\theta \quad \rightarrow \quad |G|^2 = 1 - 4\nu^2(1 - \nu^2)\sin^4(\theta/2)$$

- Stable for $\nu \le 1$; accuracy $O(\Delta t^2, \Delta x^2)$; produces dispersive errors (oscillations near discontinuities) rather than smearing.

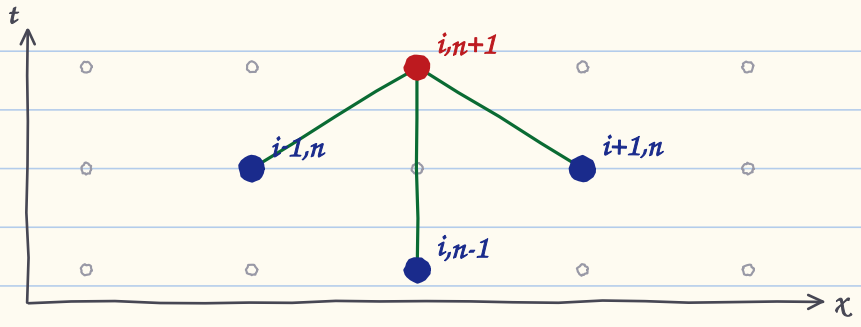
Amplification factor magnitudes ($\nu = 0.8$)



(e) Leapfrog (central in time and space)

$$u_i^{n+1} = u_i^{n-1} - \nu\,(u_{i+1}^n - u_{i-1}^n)$$

Leapfrog stencil (three time levels)



$$G^2 + 2I\nu \sin\theta \cdot G - 1 = 0 \;\rightarrow\; |G| = 1 \text{ for } \nu \le 1 \text{ (no dissipation)}$$

- Second order, non-dissipative, but needs two time levels (starter by another scheme) and has a spurious computational mode ($G_2 \approx -G_1$) causing odd–even decoupling.

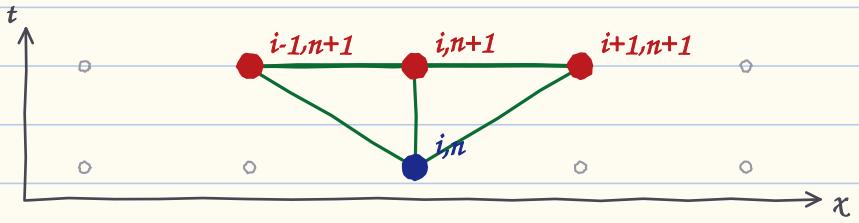
Scheme	Order	Stability	Main error
FTCS	$\Delta t, \Delta x^2$	never	growth
Upwind	$\Delta t, \Delta x$	$\nu \le 1$	diffusion
Lax	$\Delta t, \Delta x^2/\Delta t$	$\nu \le 1$	large diffusion
Lax–Wendroff	$\Delta t^2, \Delta x^2$	$\nu \le 1$	dispersion
Leapfrog	$\Delta t^2, \Delta x^2$	$\nu \le 1$	dispersion, 2 modes

3. Implicit Schemes

BTCS (fully implicit, central space)

$$u_i^{n+1} + (\nu/2)(u_{i+1}^{n+1} - u_{i-1}^{n+1}) = u_i^n$$

BTCS stencil for convection



$$G = 1/(1 + I\nu \sin\theta) \;\rightarrow\; |G|^2 = 1/(1 + \nu^2 \sin^2\theta) \le 1$$

- Unconditionally stable for any ν; tridiagonal system $(-\nu/2, 1, \nu/2)$ solved by TDMA each step; first order in time; strongly damped for large ν.

Crank–Nicolson

$$u_i^{n+1} - u_i^n = -(\nu/4)[(u_{i+1}^{n+1} - u_{i-1}^{n+1}) + (u_{i+1}^n - u_{i-1}^n)]$$

$G = (1 - I\nu \sin\theta/2)/(1 + I\nu \sin\theta/2) \;\rightarrow\; |G| = 1$ (no damping, unconditionally stable, $O(\Delta t^2, \Delta x^2)$)

Implicit upwind

$$(1 + \nu)\, u_i^{n+1} - \nu\, u_{i-1}^{n+1} = u_i^n \;\rightarrow\; \text{bidiagonal; stable for all } \nu$$

- Implicit methods remove the CFL restriction, but the wave is then not resolved accurately for large ν (phase errors). Cost: one tridiagonal solve per step. For nonlinear systems the implicit equations must be linearised.

4. Multi-step Methods

Advance the solution in two (or more) stages per time step to get high order without computing high derivatives directly.

Two-step Lax–Wendroff (Richtmyer)

$$\text{Predictor: } u_{i+1/2}^{n+1/2} = \tfrac{1}{2}(u_i^n + u_{i+1}^n) - (\nu/2)(u_{i+1}^n - u_i^n)$$

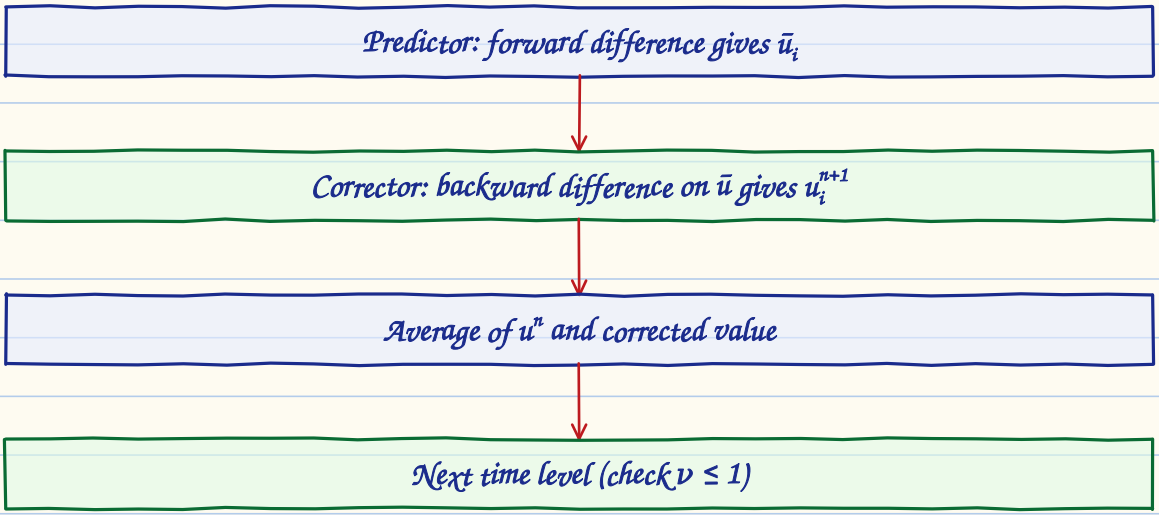
$$\text{Corrector: } u_i^{n+1} = u_i^n - \nu\,(u_{i+1/2}^{n+1/2} - u_{i-1/2}^{n+1/2})$$

Equivalent to Lax–Wendroff for linear problems but needs no Jacobian for nonlinear fluxes.

MacCormack predictor–corrector (very popular in CFD)

$$\text{Predictor: } \bar{u}_i = u_i^n - \nu\,(u_{i+1}^n - u_i^n) \quad \text{(forward difference)}$$

$$\text{Corrector: } u_i^{n+1} = \tfrac{1}{2}\,[u_i^n + \bar{u}_i - \nu\,(\bar{u}_i - \bar{u}_{i-1})] \quad \text{(backward difference)}$$



- Second order, stable for $\nu \le 1$. Alternating the direction of differences (F-B, B-F) removes bias; for linear problem it reduces to Lax–Wendroff.
- Widely used for Euler / Navier–Stokes (explicit, easy to code), with artificial viscosity added to damp oscillations at shocks.

5. Nonlinear Problems

Conservation-law form: $\partial u/\partial t + \partial f(u)/\partial x = 0$ with flux $f(u)$. Examples: Burgers $f = u^2/2$; Euler equations (vector u, vector f). Wave speed $a(u) = df/du$ depends on the solution.

- Characteristics: $dx/dt = a(u)$ — different values of u travel at different speeds -> wave steepening, characteristics converge -> shock (discontinuity) even from smooth initial data.
- Jump (Rankine–Hugoniot) condition for shock speed: $s = (f_R - f_L) / (u_R - u_L)$.
- Entropy condition selects the physical shock: $a(u_L) > s > a(u_R)$ (characteristics run into the shock).
- Use conservative difference forms $u_i^{n+1} = u_i^n - (\Delta t/\Delta x)(f_{i+1/2} - f_{i-1/2})$ so shocks have the correct speed

(Lax–Wendroff theorem). Non-conservative forms give wrong shock positions.

- Nonlinear Lax–Wendroff: $u_i^{n+1} = u_i^n - (\Delta t/2\Delta x)(f_{i+1} - f_{i-1}) + (\Delta t^2/2\Delta x^2)[A_{i+1/2}(f_{i+1} - f_i) - A_{i-1/2}(f_i - f_{i-1})]$, $A = \partial f/\partial u$.
- Stability (nonlinear): $\max|a|\,\Delta t/\Delta x \le 1$. Linearise about a local state and apply Von Neumann with frozen coefficients.
- Oscillations at shocks: add artificial viscosity $\varepsilon\,\Delta x^3\,|u_{xx}|\ldots$, use flux limiters / TVD schemes (higher order but monotone).

6. Second-order One-dimensional Wave Equation

Vibrating string, acoustic waves: $u_{tt} = c^2 u_{xx}$, $0 < x < L$, $u(x,0) = f(x)$, $u_t(x,0) = g(x)$, $u(0,t) = u(L,t) = 0$. Exact (d'Alembert): $u = \tfrac{1}{2}[f(x - ct) + f(x + ct)] + (1/2c)\int g\,ds$.

Explicit central-difference scheme

$$(u_i^{n+1} - 2u_i^n + u_i^{n-1})/\Delta t^2 = c^2\,(u_{i+1}^n - 2u_i^n + u_{i-1}^n)/\Delta x^2$$

$$\boxed{u_i^{n+1} = 2(1 - \nu^2)\,u_i^n + \nu^2(u_{i+1}^n + u_{i-1}^n) - u_i^{n-1}}$$

- Start-up: $u_i^0 = f_i$. The first step uses g: $u_i^1 = u_i^0 + \Delta t\,g_i + (\nu^2/2)(u_{i+1}^0 - 2u_i^0 + u_{i-1}^0)$.
- Von Neumann: $G + 1/G = 2(1 - 2\nu^2\sin^2(\theta/2))$ -> $|G| = 1$ if $|1 - 2\nu^2\sin^2(\theta/2)| \le 1$, i.e. $\nu = c\Delta t/\Delta x \le 1$. For $\nu = 1$ the scheme is exact on the grid.
- Second-order accurate, non-dissipative, conditionally stable (CFL).

Implicit scheme (Newmark / weighted-average)

$$(u^{n+1} - 2u^n + u^{n-1})/\Delta t^2 = c^2\,\delta_x^2/\Delta x^2\,[\beta\,u^{n+1} + (1 - 2\beta)\,u^n + \beta\,u^{n-1}]$$

- $\beta = 0$: explicit scheme above; $\beta \ge 1/4$: unconditionally stable (tridiagonal system each step); $\beta = 1/4$ is the average-acceleration (trapezoidal) method with no numerical damping.

Reduction to first-order system

Define $v = u_t$, $w = c\,u_x$: $v_t = c\,w_x$, $w_t = c\,v_x$ — a hyperbolic system $U_t + A\,U_x = 0$ with eigenvalues $\pm c$. Any first-order scheme (Lax–Wendroff, MacCormack) can be applied to U.

6A. Numerical Dissipation, Dispersion and the Modified Equation

Every scheme solves a slightly different PDE — the modified equation — obtained by expanding all terms in Taylor series about (x_i, t^n). For first-order upwind:

$$u_t + c\,u_x = (c\Delta x/2)(1 - \nu)\,u_{xx} - (c\Delta x^2/6)(1 - \nu)(1 - 2\nu)\,u_{xxx} + \ldots$$

- Even-derivative terms (u_{xx}, u_{xxxx}) -> dissipation: amplitude error, smearing of discontinuities, usually stabilising when the coefficient is positive.

- Odd-derivative terms (u_{xxx}) -> dispersion: different wavelengths travel at different speeds, giving wiggles (Gibbs-type oscillations), mostly behind a front for Lax–Wendroff.
- Exact amplification factor of the PDE: $|G_e| = 1$ and phase $-\nu\theta$. Compare numerical $|G|$ (dissipation error) and phase $\varphi = \arg G$ (dispersion error): relative phase error $\varphi/(-\nu\theta)$.

Scheme	Dominant error	Visible effect
Upwind / Lax	dissipation	profile smeared
Lax–Wendroff	dispersion	oscillations behind front
Leapfrog	dispersion	parasitic wave train
Godunov-type TVD	limited both	sharp, monotone

Godunov theorem and TVD limiters (brief)

- Godunov: a linear monotone scheme can be at most first-order accurate. To get sharp, oscillation-free shocks use non-linear schemes (TVD: total variation does not increase).
- Flux-limiter idea: blend upwind (monotone) and Lax–Wendroff (accurate): $\mathcal{F}_{i+1/2} = f^{up} + \psi(r)(f^{LW} - f^{up})$, r = ratio of consecutive gradients; e.g. minmod, van Leer, superbee limiters.

6B. Boundary Conditions for Hyperbolic Problems

- Specify conditions only where characteristics enter the domain. For $u_t + c\,u_x = 0$, $c > 0$: give u at the left (inflow) boundary; the right boundary needs no condition — use extrapolation / one-sided upwind scheme $u_N^{n+1} = u_N^n - \nu(u_N^n - u_{N-1}^n)$.
- Over-specification at outflow gives reflections and loss of well-posedness.
- Periodic problems: $u_0 \equiv u_N$ — convenient for testing schemes with Von Neumann analysis.
- For the wave equation $u_{tt} = c^2 u_{xx}$: fixed end $u = 0$, free end $u_x = 0$ (ghost node $u_{N+1} = u_{N-1}$); non-reflecting boundary $u_t + c\,u_x = 0$ at outflow.

6C. Systems of Hyperbolic Equations (1-D Euler / acoustics)

System: $U_t + A\,U_x = 0$. It is hyperbolic if the Jacobian A has real eigenvalues and a full set of eigenvectors. Diagonalise $A = R\Lambda R^{-1}$ and set $W = R^{-1}U$: $W_t + \Lambda\,W_x = 0$ — independent scalar waves (characteristic variables) with speeds λ_k.

- 1-D acoustics: $p_t + \rho c^2 u_x = 0$, $u_t + p_x/\rho = 0$: $\lambda = \pm c$. Apply the scheme to each characteristic variable, using upwind in the direction of each λ_k.
- 1-D Euler: $U = (\rho, \rho u, E)$, $\mathcal{F} = (\rho u, \rho u^2 + p, u(E + p))$; $\lambda = u, u + a, u - a$ (a = speed of sound).

CFL: $(|u| + a)\Delta t/\Delta x \le 1$.

- Applying a scalar scheme componentwise (e.g. Lax–Wendroff with matrix A) gives schemes for systems; stability requires $\rho(A)\Delta t/\Delta x \le 1$ where $\rho(A) = \max|\lambda|$.

6D. Two-dimensional Convection

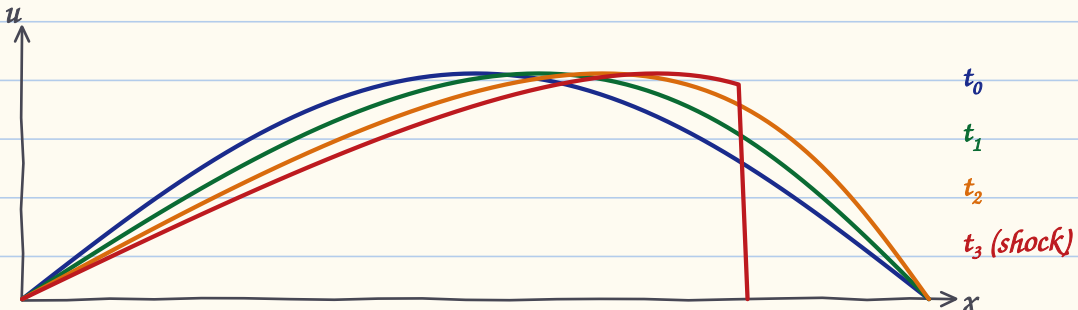
$u_t + a\,u_x + b\,u_y = 0$. Explicit upwind is stable for $|\nu_x| + |\nu_y| \le 1$ ($\nu_x = a\Delta t/\Delta x$, $\nu_y = b\Delta t/\Delta y$). Alternatives: fractional steps (solve x-sweep then y-sweep), 2-D Lax–Wendroff, MacCormack applied in each direction; implicit ADI for large time steps.

7. Burgers Equation

Simplest model that combines nonlinear convection and diffusion — a prototype for the Navier–Stokes momentum equation:

$$\partial u/\partial t + u\,\partial u/\partial x = \nu\,\partial^2 u/\partial x^2 \quad \text{(viscous Burgers)}$$

- $\nu = 0$: inviscid Burgers equation, hyperbolic, develops shocks ($f = u^2/2$, shock speed $s = (u_L + u_R)/2$).
- $\nu > 0$: parabolic; viscosity balances steepening -> smooth shock layer of thickness ~ $\nu/\Delta u$.
- Exact solution via Cole–Hopf: $u = -2\nu\,\varphi_x/\varphi$ transforms Burgers into the heat equation $\varphi_t = \nu\,\varphi_{xx}$.



steepening wave -> shock

Explicit scheme (FTCS)

$$u_i^{n+1} = u_i^n - (\Delta t/2\Delta x)\,u_i^n(u_{i+1}^n - u_{i-1}^n) + (\nu\Delta t/\Delta x^2)(u_{i+1}^n - 2u_i^n + u_{i-1}^n)$$

- Stability (approximate, Von Neumann with frozen u): $2\nu\Delta t/\Delta x^2 \le 1$ and $(\Delta t/\nu)\,u^2 \le 2$ i.e. $\Delta t \le \min[\Delta x^2/2\nu\ ,\ 2\nu/u^2_{max}]$.
- Cell Reynolds number $Re_\Delta = u\Delta x/\nu \le 2$ required to avoid oscillations with central convection.
- Upwind convection alternative: $-(\Delta t/\Delta x)\,u_i^n(u_i^n - u_{i-1}^n)$ (for $u > 0$): more stable, more diffusive; conservative form $-(\Delta t/2\Delta x)(F_{i+1/2} - F_{i-1/2})$, $F = u^2/2$.

Implicit scheme

Fully implicit (BTCS) with linearised convection u_i^n (lagged coefficient):

$$-(r + C_i)\,u_{i-1}^{n+1} + (1 + 2r)\,u_i^{n+1} + (C_i - r)\,u_{i+1}^{n+1} = u_i^n$$

where $r = \nu\Delta t/\Delta x^2$ and $C_i = u_i^n\Delta t/2\Delta x$. Tridiagonal system -> TDMA each time step; unconditionally stable for the linearised form, large Δt allowed.

- Fully non-linear implicit ($u^{n+1}u_x^{n+1}$): solve by Newton iteration with a tridiagonal Jacobian, or by Picard (lagging) iteration within the time step.
- Crank–Nicolson version: average convective and diffusive terms at n and n+1; second order in time.

8. Runge–Kutta Methods (time integration)

Method of lines: discretise space first, $du/dt = \mathcal{F}(u)$ (a system of ODEs, e.g. $\mathcal{F} = -u\,u_x + \nu\,u_{xx}$ for Burgers), then integrate in time with a high-order explicit multistage scheme.

Second-order RK (Heun)

$$k_1 = \Delta t\,\mathcal{F}(u^n);\quad k_2 = \Delta t\,\mathcal{F}(u^n + k_1);\quad u^{n+1} = u^n + \tfrac{1}{2}(k_1 + k_2)$$

Classical fourth-order RK

$$u^{n+1} = u^n + (1/6)(k_1 + 2k_2 + 2k_3 + k_4)$$

$$k_1 = \Delta t\mathcal{F}(u^n),\ k_2 = \Delta t\mathcal{F}(u^n + k_1/2),\ k_3 = \Delta t\mathcal{F}(u^n + k_2/2),\ k_4 = \Delta t\mathcal{F}(u^n + k_3)$$

Third-order TVD (Shu–Osher) — preferred for shocks

$$u^{(1)} = u^n + \Delta t\mathcal{F}(u^n);\quad u^{(2)} = \tfrac{3}{4}u^n + \tfrac{1}{4}[u^{(1)} + \Delta t\mathcal{F}(u^{(1)})];\quad u^{n+1} = \tfrac{1}{3}u^n + \tfrac{2}{3}[u^{(2)} + \Delta t\mathcal{F}(u^{(2)})]$$

- RK4 has a large stability region on the imaginary axis ($|\lambda\Delta t| \le 2.8$) so it works with central space differences for convection; RK2 is unstable for purely imaginary eigenvalues (central FTCS equivalent).
- Advantages: high-order in time, self-starting, easy to code, no linear solve. Disadvantage: several $\mathcal{F}$ evaluations per step; explicit time-step limit (CFL).

9. Worked Examples

Example 1 — Upwind scheme, step by step

Data: $c = 1$, $\Delta x = 0.1$, $\Delta t = 0.05$ -> $\nu = 0.5$. Initial: $u = 1$ for $x \le 0.2$, $u = 0$ beyond (step). Inflow $u = 1$.

Update $u_i^{n+1} = u_i^n - 0.5(u_i^n - u_{i-1}^n) = 0.5u_i^n + 0.5u_{i-1}^n$.

Nodes $x = 0.1, 0.2, 0.3, 0.4$: initial 1, 1, 0, 0. After step 1: 1, 1, 0.5, 0. After step 2: 1, 1, 0.75, 0.25.

The sharp step is smeared over several cells — numerical diffusion; exact solution would be a sharp step moving with speed 1.

Example 2 — Stability check by Von Neumann (upwind)

$G = 1 - \nu + \nu e^{-I\theta}$ -> $|G|^2 = (1 - \nu + \nu \cos\theta)^2 + \nu^2 \sin^2\theta = 1 - 2\nu(1 - \nu)(1 - \cos\theta)$.

$|G|^2 \le 1$ requires $\nu(1 - \nu) \ge 0$, i.e. $0 \le \nu \le 1$ √. For $\nu = 1.2$: $|G|^2 = 1 + 0.48(1 - \cos\theta) > 1$ — unstable.

Example 3 — Burgers explicit step

$\nu = 0.01$, $\Delta x = 0.1$, $\Delta t = 0.01$, $u_{i-1} = 1.0$, $u_i = 0.8$, $u_{i+1} = 0.5$.

Convective term: $-(0.01/0.2)(0.8)(0.5 - 1.0) = +0.02$. Diffusive term: $(0.01 \cdot 0.01/0.01)(0.5 - 1.6 + 1.0) = -0.001$.

$u_i^{n+1} = 0.8 + 0.02 - 0.001 = 0.819$. Check: $2\nu\Delta t/\Delta x^2 = 0.02 \le 1$ √, $Re_\Delta = 8 > 2$ — central convection may oscillate; reduce Δx or use upwind.

Stability regions of Runge–Kutta methods

For $du/dt = \lambda u$ with $z = \lambda\Delta t$, RK stability requires $|R(z)| \le 1$, where $R(z) = 1 + z + z^2/2$ (RK2), $R(z) = 1 + z + z^2/2 + z^3/6 + z^4/24$ (RK4). Along the negative real axis: RK2 $z \ge -2$, RK4 $z \ge -2.78$; along the imaginary axis RK4: $|\text{Im } z| \le 2.83$ -> usable for convection with central differences when $\nu \le 2.8/(\max|\sin\theta|) \approx 2.8$.

10. Short Answer Questions

Q1. What is a hyperbolic PDE? -> $B^2 - 4AC > 0$; has real characteristics; signals travel at finite speed (wave equation, supersonic flow).

Q2. State the CFL condition. -> $\nu = c\Delta t/\Delta x \le 1$: numerical domain of dependence must contain the

physical one.

Q3. Why upwind differencing? -> information moves along characteristics, so the difference must use upstream points; yields stability and monotonicity at the cost of numerical diffusion.

Q4. What is numerical dissipation? -> amplitude damping from even-order truncation terms ($|G| < 1$); smears discontinuities.

Q5. What is numerical dispersion? -> phase error from odd-order truncation terms; different wavelengths move at different speeds.

Q6. Predictor–corrector advantage? -> second-order accuracy, explicit, no Jacobian matrices; MacCormack is the standard example.

Q7. Why conservative form for nonlinear laws? -> ensures correct shock speed through Rankine–Hugoniot jump conditions (Lax–Wendroff theorem).

Q8. Cell Reynolds number? -> $Re_\Delta = u\Delta x/\nu$; central scheme for Burgers/convection–diffusion stays oscillation-free only for $Re_\Delta \le 2$.

Q9. Leapfrog drawback? -> spurious computational mode and need for two starting levels; no damping to control it.

Q10. Why RK methods? -> high order in time, self-starting, easy; combined with method of lines for any spatial discretisation.

Comparison chart of schemes for $u_t + c\,u_x = 0$

Scheme	Explicit?	Levels	ν limit	Accuracy
Upwind	yes	2	1	1st
Lax	yes	2	1	1st
Lax–Wendroff	yes	2	1	2nd
Leapfrog	yes	3	1	2nd
MacCormack	yes	2 (2-stage)	1	2nd
BTCS	no	2	none	1st
Crank–Nicolson	no	2	none	2nd

Quick revision — Unit II

- Hyperbolic: finite speed, characteristics $dx/dt = c$; CFL $\nu = c\Delta t/\Delta x \le 1$.
- FTCS unstable; upwind (1st order, diffusive), Lax–Wendroff (2nd order, dispersive), leapfrog (non-dissipative), MacCormack (2-step).
- Implicit BTCS / Crank–Nicolson: unconditionally stable, tridiagonal.
- Wave equation $u_{tt} = c^2 u_{xx}$: explicit stable for $\nu \le 1$; Newmark $\beta \ge 1/4$ unconditionally stable.
- Burgers: nonlinear convection + diffusion; $Re_\Delta \le 2$ for central; RK2/RK4/TVD-RK3 for time integration.

Important Questions

1. Explain characteristics and the CFL condition. Derive the stability limits of upwind and Lax–Wendroff by Von Neumann analysis.
2. Why is FTCS unstable for $u_t + c\,u_x = 0$? How does the Lax scheme remove this?
3. Explain MacCormack and two-step Lax–Wendroff schemes.
4. Derive the explicit scheme for the wave equation $u_{tt} = c^2 u_{xx}$ and discuss its stability.
5. Write explicit and implicit schemes for Burgers equation; explain Runge–Kutta (RK4).